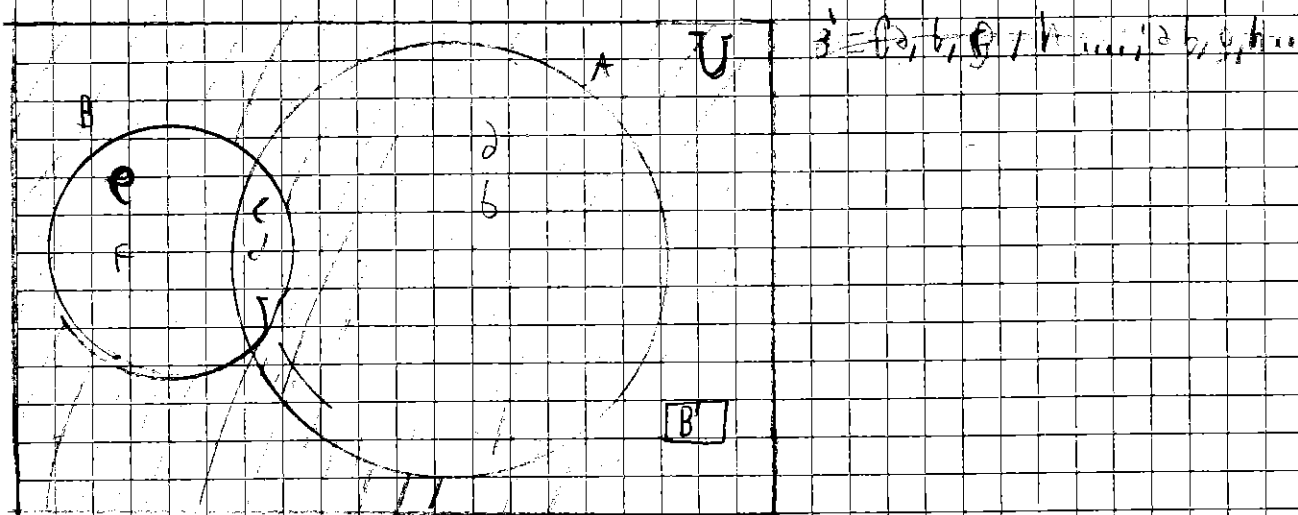


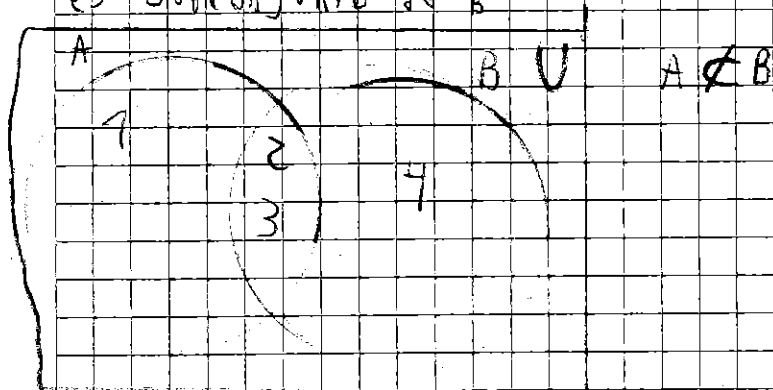
5) $A = \{a, b, c, d\}$ y $B = \{c, d, e, f\}$, determina el conjunto complementario de B con respecto a A



$$B' = \{a, b, c, d, \dots, a, b, g, h\}$$

$$B' = \{a, b, \dots, a, b, \dots \in U \wedge a, b, \dots \notin B\}$$

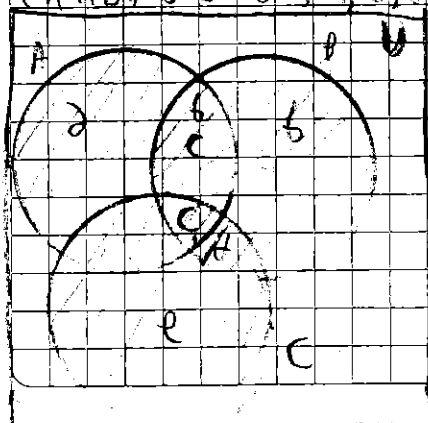
2) Dados los conjuntos $A = \{1, 2, 3\}$ y $B = \{2, 3, 4\}$, verifica que A es subconjunto de B



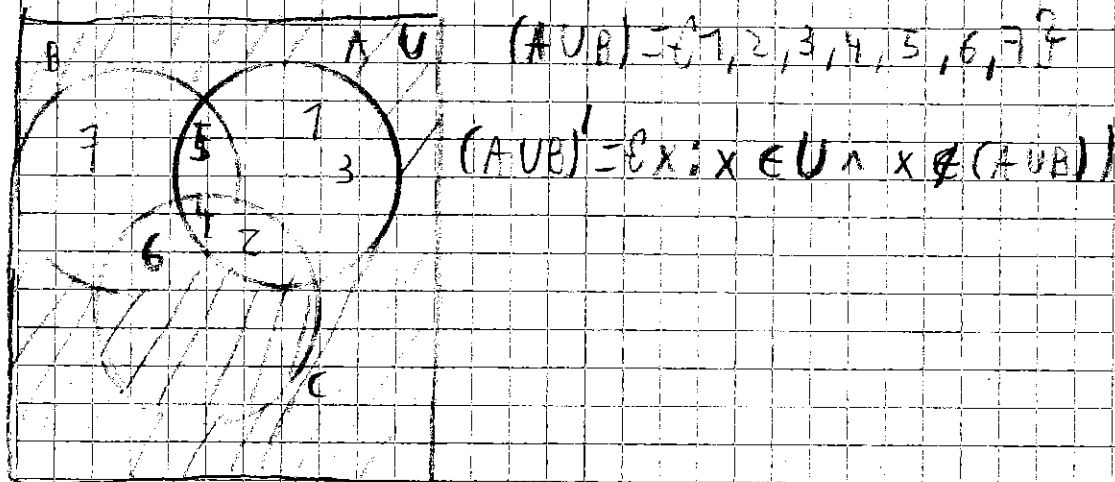
3) Si U es el conjunto universo y $A = \{a, b, c\}$, $B = \{b, c, d\}$, $C = \{c, d, e, f, g\}$

$$A \cap B = \{b, c, d\}$$

$$(A \cap B) \cup C = \{a, b, c, d, e, f, g\}$$



5) Si $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$, y $C = \{2, 7, 6\}$ determine el conjunto con complemento de $(A \cup B)$ con respecto a U y C

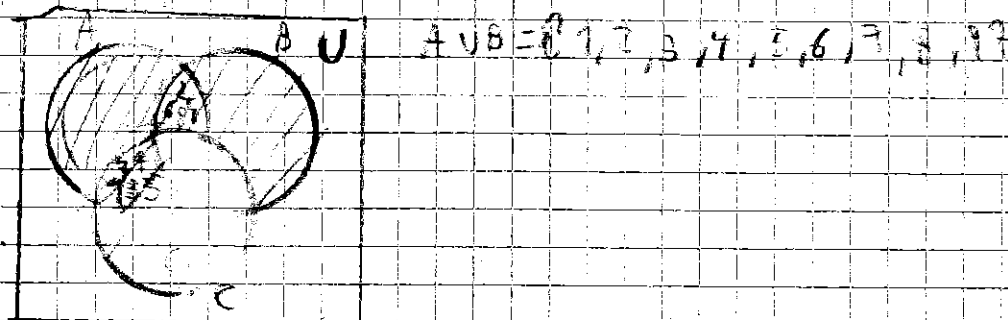


5) Si $A = \{x | x \text{ es un número primo menor que } 10\}$ y $B = \{x | x \text{ es un número par}\}$, encuentra $A \cap B$

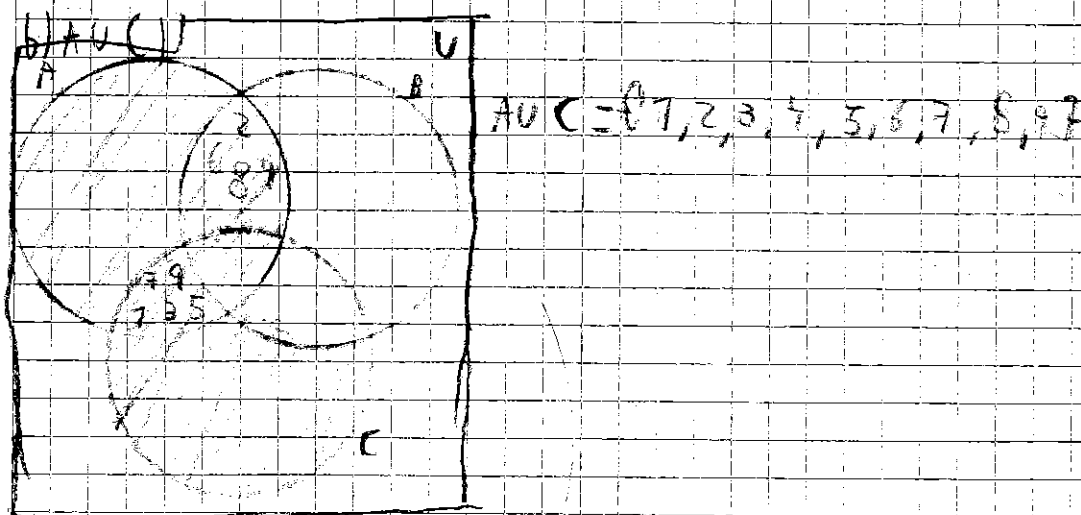
$A = \{2, 3, 5, 7\}$ $B = \{x | 2x = m\}$ $m \text{ es un número par}$
 $A \cap B = \{2\}$

6) Sean $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $B = \{2, 4, 6, 8\}$, $C = \{1, 3, 5, 7, 9\}$, $U = \mathbb{N}$ (el conjunto de los números naturales), realiza las siguientes operaciones

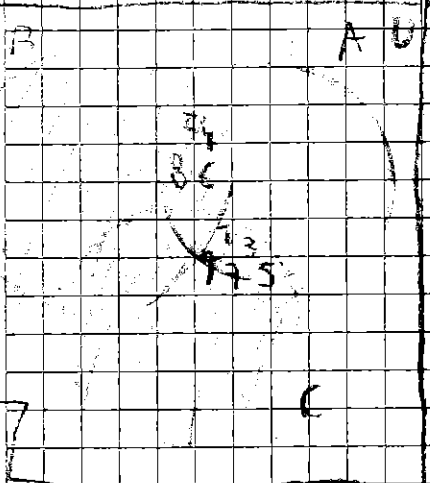
a) $A \cup B$



b) $A \cup C$

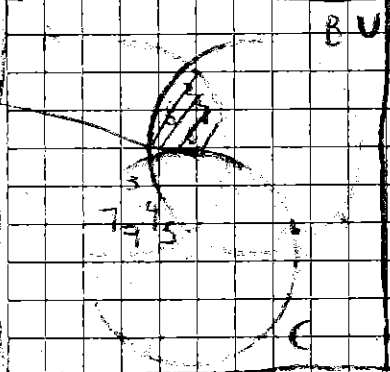


c) $B \cup C$



$$B \cup C = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

d) $A \cap B$



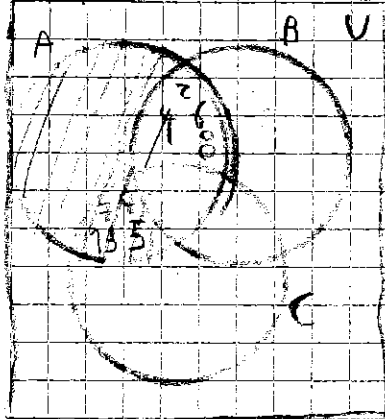
$$A \cap B = \{2, 8, 9, 6\}$$

e) $B \cap C$



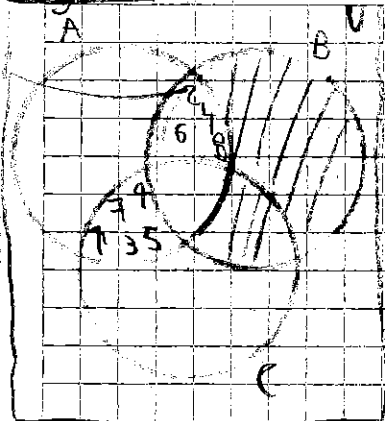
$$B \cap C = \{4\}$$

1) A-B



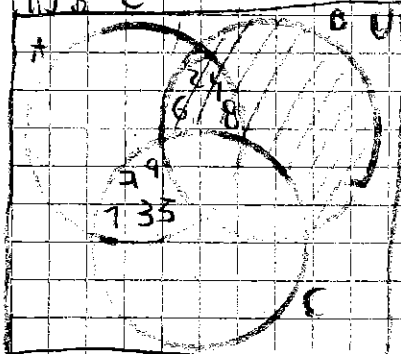
$$A-B = \{1, 3, 5\}$$

2) B-A



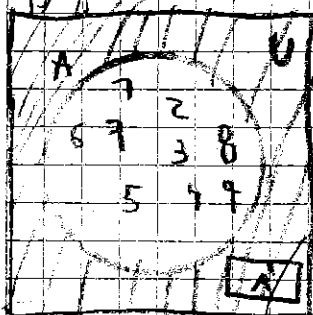
$$B-A = \{\emptyset\}$$

3) B-C



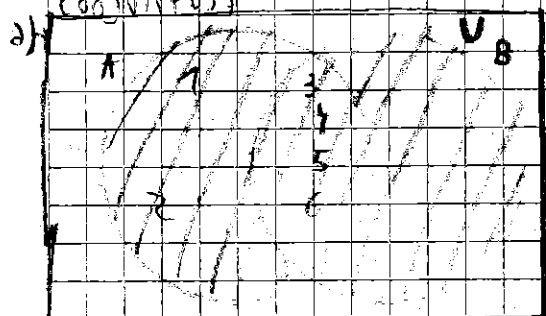
$$B-C = \{2, 4, 6, 8\}$$

4) A'

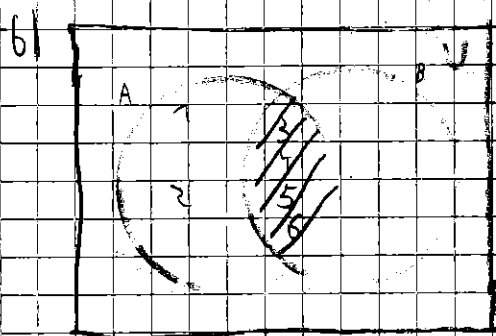


$$A' = \{x \mid x \in U \wedge x \notin A\}$$

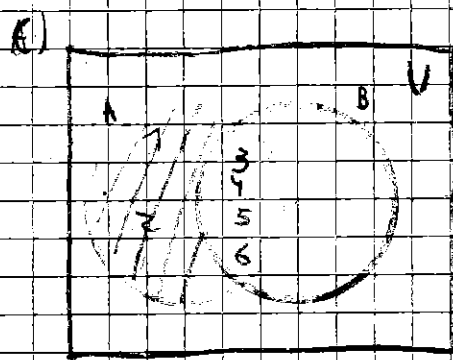
7) Dados los conjuntos $A = \{1, 2, 3, 4, 5, 6\}$ y $B = \{3, 4, 5, 6\}$, determina:
 a) $A \cup B$ (unión de conjuntos) ~~$A \cup B = \{1, 2, 3, 4, 5, 6\}$~~ b) $A \cap B$ (intersección de conjuntos) c) $A - B$ (diferencia de conjuntos) d) $B - A$ (diferencia de conjuntos)



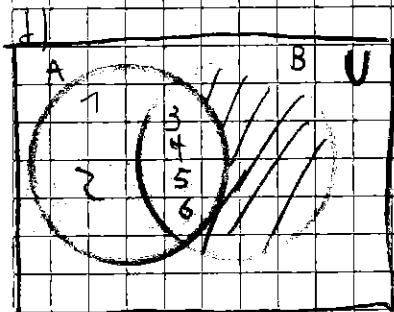
$$A \cup B = \{1, 2, 3, 4, 5, 6\}$$



$$A \cap B = \{3, 4, 5, 6\}$$

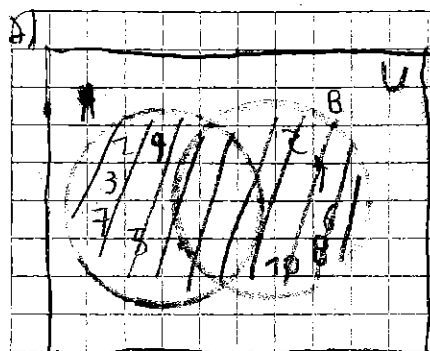


$$A - B = \{1, 2\}$$

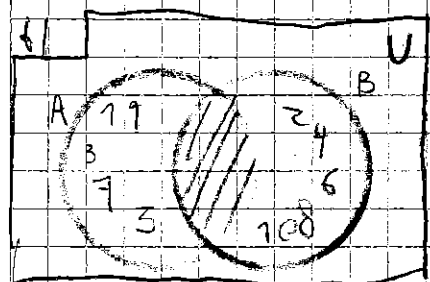


$$B - A = \{3, 4, 5, 6\}$$

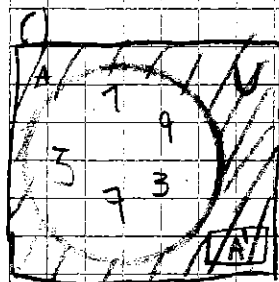
8) Si el conjunto universo $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, y $A = \{1, 3, 5, 7, 9\}$ y $B = \{2, 4, 6, 8, 10\}$, encuentra: a) $A \cup B$ b) $A \cap B$ c) A' (complemento de A) d) B' (complemento de B)



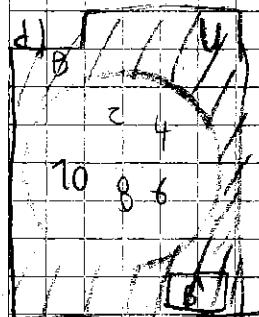
$$A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$



$$A \cap B = \{\emptyset\}$$



$$A' = \{2, 4, 6, 8, 10\}$$



$$B' = \{1, 3, 5, 7, 9\}$$

7) Considere el conjunto $A = \{1, 2, 3, 4\}$ y $B = \{3, 4, 5, 6\}$. encuentre la cardinalidad de: a) $A \cap B$ b) $A \cup B$ c) $A \cap B$

a) $n(A) = 4$

b) $n(B) = 4$

c) $n(A \cup B) = 6$

d) $n(A \cap B) = 2$

10) Dados los conjuntos $A = \{a, b, c, d, e\}$ y $B = \{c, d, e, f, g\}$, verifique si las siguientes afirmaciones son verdaderas o falsas justifique su respuesta: a) $A \subseteq B$ (A es subconjunto de B) b) $B \subseteq A$ (B es subconjunto de A) c) $A \subseteq B \cup A$

a) $A \subseteq B$ es falsa, no todos los elementos de A están en B

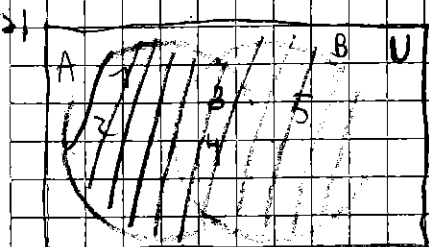
b) $B \subseteq A$ es falsa, no todos los elementos de B están en A

c) $A \subset B$ es falsa, A no es subconjunto de B

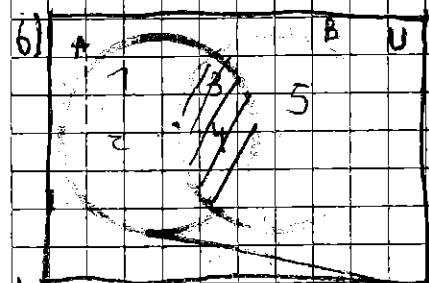
d) $B \subset A$ es falsa, B no es subconjunto de A

77) Dados los conjuntos $A = \{1, 2, 3, 4\}$ y $B = \{3, 4, 5\}$ determina los conjuntos resultantes de las siguientes operaciones: a) $A \cup B$ (unión)

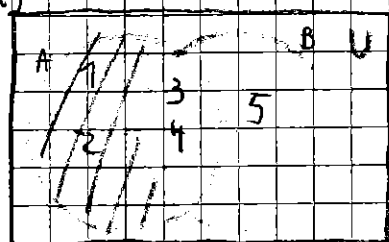
b) $A \cap B$ (intersección) c) $A - B$ (diferencia) d) $B - A$ (diferencia)



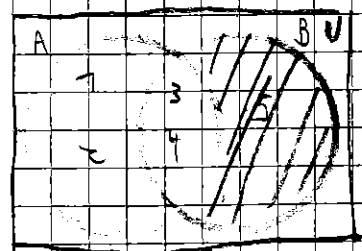
$$A \cup B = \{1, 2, 3, 4, 5\}$$



$$A \cap B = \{3, 4\}$$



$$A - B = \{1, 2\}$$



$$B - A = \{5\}$$

12) Dados los conjuntos $A = \{1, 2, 3, 4\}$ y $B = \{3, 4, 5\}$, determina si las siguientes afirmaciones son verdaderas o falsas, justifica tu respuesta: a) $2 \in A$ b) $5 \notin A$ c) $A \subseteq B$ d) $B \subseteq A$.

a) $2 \in A$: Verdadero, el 2 es elemento del conjunto A

b) $5 \notin A$: Verdadero, el 5 no es elemento del conjunto A

c) $A \subseteq B$: Falso, no todos los elementos de A están en B

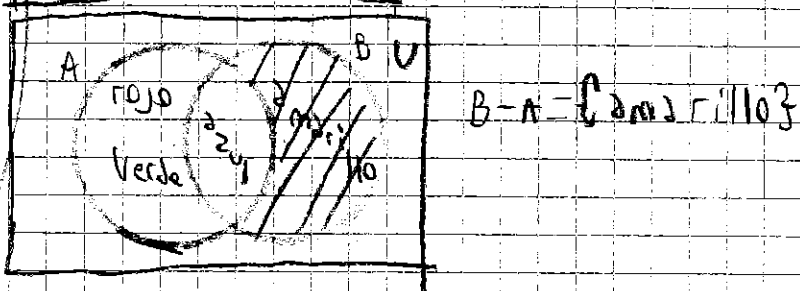
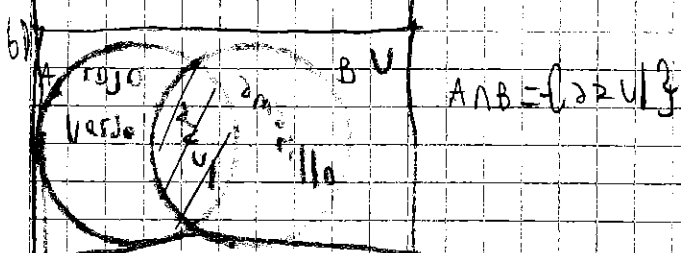
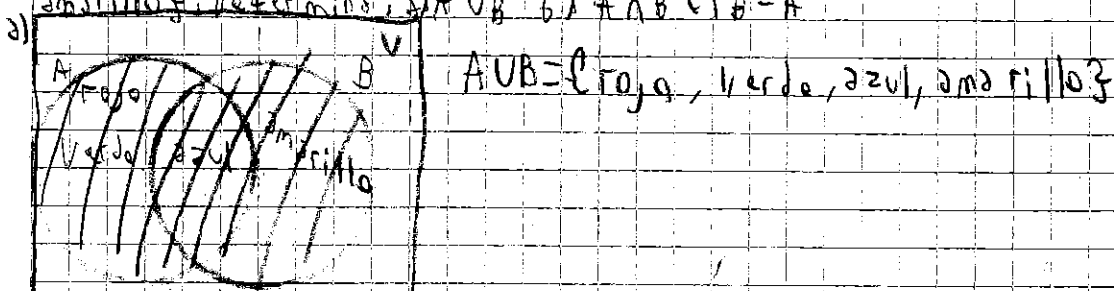
d) $B \subseteq A$: Falso, no todos los elementos de B están en A

13) Considere los conjuntos $A = \{x \mid x \text{ es un número entero positivo menor que } 5\}$ y $B = \{x \mid x \text{ es un número primo menor que } 10\}$. Encuentra: a) $A \cup B$ b) $A \cap B$

a) $A \cup B = \{1, 2, 3, 4, 5, 7\}$

b) $A \cap B = \{2, 3\}$

14) Considere los conjuntos $A = \{\text{rojo, verde, azul}\}$ y $B = \{\text{azul, amarillo}\}$. Determina: a) $A \cup B$ b) $A \cap B$ c) $B - A$



15) Resuelve los siguientes problemas y marca con una cruz la respuesta correcta:

- a) Se hizo una encuesta a 50 estudiantes sobre sus preferencias respecto a dos libros A y B. Se observó que los que leen ambos libros son el doble de los que leen solo A, el triple de los que leen solo B y el cuádruplo de los que no leen ninguno de los dos libros. ¿Cuántas personas leen la revista A?

4

UNIVERSIDAD TECNOLÓGICA NACIONAL – FACULTAD REGIONAL MENDOZA – SEMINARIO DE INGRESO 2024 -

- a. 24
b. 32
c. 36
d. NRC ☒

- b) Se realizó una encuesta a todos los ingresantes a ingeniería sobre la práctica de algún deporte. El 60% de los estudiantes practica fútbol; el 50% tenis, el 40% de los que practican tenis también practican fútbol ¿Qué porcentaje de los estudiantes no realiza ninguno de estos deportes?

- a. 10%
b. 30% ☒
c. 15%
d. NRC

) Coloca $\in$, $\notin$, $\subset$, $\not\subset$ según corresponda:

$-3 \dots^E \dots Z$	$\sqrt{-9} \dots^E \dots R$	$1.3333 \dots^{\cancel{E}} \dots I$
$-1/3 \dots^E \dots Q$	$I \dots^{\cancel{E}} \dots Q$	$(\sqrt{2} + \frac{4}{3}) \dots^E \dots R$
$\sqrt{2} \dots^{\cancel{E}} \dots Q$	$I \dots^{\cancel{E}} \dots R$	$0 \dots^E \dots Z$
$\sqrt{3} \dots^E \dots R$	$1 - \sqrt{3} \dots^E \dots I$	$\sqrt{-25} \dots^E \dots R$

18) Tacha los números que no correspondan a la clasificación:

Naturales: 0 ; ~~1~~ ; 41 ; ~~0,8~~ ; 2 ; ; ~~1,131133111~~.....

Enteros: -4 ; 25 ; 0 ; ~~1/2~~ ; ~~0,2~~ ; 47 ; ~~2,6~~ ; ~~-1,5~~.

Racionales: ~~4~~ ; 2/5 ; ~~0~~ ; 2,23 ; 1, 8 ; ~~-5~~² ; ~~2~~ ; -1,5

Irracionales: ~~4~~ ; ~~1/3~~ ; ~~2,8~~ ; ; ~~7,2~~ ; 7,212200148.... ; $2\sqrt{2}$; $-\sqrt[3]{5}$, $-2\sqrt{2}$;

19) Completa:

- Un ejemplo de un número racional no entero es $\frac{1}{2}$
- Un ejemplo de un número real no racional es $\frac{1}{2}$
- Un ejemplo de un número real no irracional $\frac{2}{2}$
- Un ejemplo de un número entero no natural $-1/3$

76) Dados los siguientes números reales, ordénalos de menor a mayor

$-1,4777\dots$; $-1,5\sqrt{2}$; 2 ; $1,9999\dots$; $(\sqrt{2}-0,2)$; $1,4444\dots$; $-1,4444$

$$-1,5\sqrt{2} < -1,4777\dots < -1,4444\dots < \sqrt{2}-0,2 < 1,4444\dots < 1,9999\dots < 2$$

20) Resuelve

$$-4 + \frac{3}{4} - \frac{1}{2} \cdot 5 + \frac{5}{8} \cdot \left(-\frac{2}{15}\right) - \frac{3}{2} \cdot \left(-\frac{1}{2} + \frac{1}{2} \cdot (-4)\right)$$

$$-\frac{13}{4} - \frac{5}{2} - \frac{1}{12} - \frac{3}{2} \cdot \left(-\frac{1}{2} - 2\right)$$

$$-\frac{35}{6} - \frac{3}{2} \cdot \left(-\frac{5}{2}\right)$$

$$-\frac{25}{12}$$

$$R+A: -\frac{25}{12}$$

$$b) -\left(-\frac{2}{3}\right)^2 + \left(\frac{7}{2}\right)^3 \cdot (-1) - \frac{2}{3} \cdot (-2)^3 + \frac{11}{2} - \left(\frac{1}{6}\right)^2 \cdot 3^{-1}$$

$$-\left(\frac{4}{9}\right) + \left(\frac{7}{8}\right) \cdot (-1) - \frac{2}{3} \cdot (-8) + \frac{11}{2} - \left(\frac{1}{36}\right) \cdot \frac{1}{3}$$

$$-\frac{4}{9} + \frac{7}{8} + \frac{16}{3} + \frac{11}{2} - \frac{1}{108}$$

$$\frac{8777}{1080}$$

$$c) \frac{(\sqrt{6} \cdot \sqrt[4]{12})^3}{18^{\frac{1}{2}}}$$

$$\frac{(\sqrt{6} \cdot \sqrt[4]{12})^3}{\sqrt{18}}$$

$$\frac{(\sqrt{6})^3 \cdot (\sqrt[4]{12})^3}{3\sqrt{2}}$$

$$\frac{\sqrt{76} \cdot \sqrt[4]{1728}}{\sqrt{18}}$$

$$\frac{\sqrt{2^3 \cdot 19} \cdot \sqrt[4]{2^6 \cdot 3^3}}{\sqrt{2 \cdot 3^2}}$$

$$\frac{2 \cdot 3 \sqrt{2 \cdot 3} \cdot 2 \cdot \sqrt[4]{2^2 \cdot 3}}{3\sqrt{2}}$$

$$\frac{6\sqrt{6} \cdot 2\sqrt[4]{768}}{3\sqrt{2}}$$

$$\frac{6 \cdot 2 \sqrt[4]{6^4 \cdot 128}}{3\sqrt{2}}$$

$$\frac{12 \sqrt{3888}}{3\sqrt{2}}$$

$$\frac{12 \cdot 6 \sqrt{3}}{3\sqrt{2}}$$

$$\frac{72\sqrt{3}}{3\sqrt{2}}$$

$$\frac{24\sqrt{3}}{\sqrt{2}}$$

$$R+A = \frac{24\sqrt{3}}{\sqrt{2}}$$

$$d) -100^{\frac{1}{2}}$$

$$\sqrt[3]{10}$$

$$\sqrt[6]{1000}$$

$$-10$$

$$\sqrt[3]{10} \cdot \sqrt[6]{1000}$$

$$-10$$

$$\sqrt[3]{10} \cdot \sqrt[6]{1000}$$

$$-10$$

$$\sqrt[3]{\sqrt{10^2} \cdot 1000}$$

$$-10$$

$$\sqrt[6]{100000}$$

$$\frac{-10}{\sqrt[6]{10^8}} \cdot \frac{\sqrt[6]{10}}{\sqrt[6]{10}}$$

$$\frac{-10 \sqrt[6]{10}}{10}$$

$$R+2: -\sqrt[6]{10}$$

$$g) 14 \frac{3}{4} \cdot \left(\frac{3}{4}\right)^2$$

$$21 \frac{6}{8} \cdot \frac{9}{8}$$

$$14 \frac{3}{4} \cdot \frac{9}{8}$$

$$21 \frac{6}{8} \cdot \frac{9}{8}$$

$$21 \frac{6}{8} \cdot \frac{9}{8}$$

$$e) \frac{10^{2n+1}}{10^{n+1}}$$

$$\frac{10^{2n+1}}{10^{n+1}}$$

$$\frac{10^n \cdot 10^n \cdot 10}{10^n \cdot 10}$$

$$10^n$$

$$R+2: 10^n$$

$$f) \frac{x^{-3} y^4}{x^4 y^{-3}}$$

$$\frac{x^{-7} y^7}{x^7}$$

$$R+2: \frac{y^7}{x^7}$$

$$h) \left(\frac{a^{-1}-b^{-1}}{a^{-1}+b^{-1}}\right)^{-1}$$

$$\left(\frac{1}{a} - \frac{1}{b}\right)^{-1} \cdot \left(\frac{1}{a} + \frac{1}{b}\right)^{-1}$$

$$\left(\frac{b+a}{ab}\right)^{-1} \cdot \left(\frac{b+a}{ab}\right)^{-1}$$

$$\left(\frac{ab}{b+a}\right) \cdot \left(\frac{ab}{b+a}\right)$$

$$\left(\frac{(ab) \cdot (ab)}{(b+a) \cdot (b+a)}\right)$$

$$\frac{(ab)^2}{(b+a)^2 - a^2 - b^2}$$

$$R+2: \frac{a^2 b^2}{b^2 - a^2}$$

$$1. \sqrt{\frac{2}{2}} - \sqrt{\frac{1}{22}} - \sqrt{\frac{2}{2}}$$

$$\frac{\sqrt{2}}{\sqrt{2}} - \frac{\sqrt{1}}{\sqrt{22}} - \frac{\sqrt{2}}{\sqrt{2}}$$

$$\frac{\sqrt{2} \cdot \sqrt{2}}{\sqrt{2}} - \frac{\sqrt{22}}{\sqrt{22}} - \frac{\sqrt{2} \cdot \sqrt{2}}{\sqrt{2}}$$

$$\frac{\sqrt{22}}{\sqrt{22}} - \frac{\sqrt{22}}{\sqrt{22}} - \frac{\sqrt{22}}{\sqrt{22}}$$

$$\sqrt{22} \left(\frac{1}{2} - \frac{1}{22} - \frac{1}{2} \right)$$

$$\sqrt{22} \left(\frac{2-2}{22} - \frac{1}{22} \right)$$

$$R+2: \sqrt{22} \left(\frac{2-3}{22} \right)$$

$$K. \left(1 - \frac{3}{2} \right)^2 \cdot (-2)^3$$

$$\frac{-5+3}{-2 \cdot 3}$$

$$\left(-\frac{1}{2} \right)^2 \cdot (-8) \quad \left(\frac{-8}{4} \right) \cdot \frac{-2}{-6}$$

$$\frac{-2}{-6}$$

$$\frac{1}{-6}$$

$$\frac{4}{-6}$$

$$\frac{-2}{-6}$$

$$\frac{-8}{-6}$$

$$\frac{-4}{-6}$$

$$\frac{-2}{-6}$$

$$R+2: -6$$

$$1. \left(1 - \frac{1}{\sqrt{2}} \right)^{-1}$$

$$\left(\frac{\sqrt{2}-1}{\sqrt{2}} \right)^{-1}$$

$$\left(\frac{(\sqrt{2}-1)\sqrt{2}}{2} \right)^{-1}$$

$$\left(\frac{2-1\sqrt{2}}{2} \right)^{-1}$$

$$R+2: \frac{2}{2-1\sqrt{2}}$$

$$11. \left(\frac{1}{3} - 1 \right)^2 \cdot (-12)$$

$$\sqrt{\frac{16}{25}} + 1$$

$$\left(\frac{-2}{3} \right)^2 \cdot (-12)$$

$$\sqrt{\frac{36}{25}}$$

$$\left(\frac{4}{6} \right) \cdot (-12)$$

$$\sqrt{\frac{36}{25}}$$

$$\left(\frac{-48}{6} \right)$$

$$\sqrt{\frac{36}{25}}$$

$$\left(\frac{-48}{6} \right)$$

$$\frac{6}{5}$$

$$R+2: \frac{-20}{3}$$

$$\begin{aligned}
 m) & \frac{\sqrt{\frac{16}{25}} \cdot \left(\frac{1}{5}\right)^2}{\left(3 + \frac{1}{9}\right)^2} \\
 & \frac{\frac{4}{5} \cdot \left(\frac{1}{25}\right)}{\left(\frac{20}{3}\right)^2} \\
 & \frac{\frac{4}{725}}{\frac{100}{9}}
 \end{aligned}$$

$$R + s: \frac{76}{95}$$

$$\begin{aligned}
 n) & \frac{\sqrt{\left(\frac{2}{3}\right)^3 \cdot \left(\frac{2}{3}\right)^{-9}} + \frac{5}{4}}{1 - \left(\frac{1}{4}\right)^{-2}} \\
 & \frac{\sqrt{\left(\frac{2}{3}\right)^{-6}} + \frac{5}{4}}{1 - \left(\frac{1}{4}\right)^{-2}} \\
 & \frac{\sqrt{\left(\frac{3}{2}\right)^6} + \frac{5}{4}}{1 - \left(\frac{1}{16}\right)} \\
 & \frac{\left(\frac{3}{2}\right)^2 + \frac{5}{4}}{-75}
 \end{aligned}$$

$$\begin{aligned}
 & \frac{\frac{9}{4} + \frac{5}{4}}{-75} \\
 & \frac{\frac{7}{2}}{-75} \\
 & \frac{-7}{-30}
 \end{aligned}$$

$$\begin{aligned}
 o) & \frac{\frac{7}{10} - \frac{1}{5} + \frac{1}{4}}{4 - \frac{1}{2} - 2 - \frac{1}{9}} \\
 & \frac{\frac{35 - 10}{50} + \frac{1}{4}}{\frac{2 - 1}{2} + \frac{8 - 1}{4}} \\
 & \frac{\frac{1}{2} + \frac{1}{4}}{\frac{1}{2} + \frac{7}{4}} \\
 & \frac{1 + \frac{1}{2}}{\frac{8}{7}}
 \end{aligned}$$

$$p) \frac{2 - \frac{7}{3} - \frac{2}{3}}{2 + \frac{1}{3}} + \frac{7 - \frac{7}{2}}{7 + \frac{1}{2}} - \frac{1}{7 - \frac{3}{4}}$$

$$\frac{2 - 1}{\frac{7}{3}} + \frac{2 - 7}{\frac{2 + 7}{2}} - \frac{1}{\frac{4 - 3}{4}}$$

$$\frac{1}{\frac{7}{3}} + \frac{-5}{\frac{2}{3}} - \frac{1}{\frac{1}{4}}$$

$$\frac{3}{7} - \frac{5}{3} - 4$$

$$R + 0: -100 \\ 21$$

$$q) \left(\frac{2}{3}\right)^{-2} \cdot \left(\frac{3}{2}\right)^{\frac{2}{3}}$$

$$\frac{2^{-6} \cdot 3}{2^5 \cdot 3^{\frac{1}{3}}}$$

$$\frac{7 \cdot 3}{2^6 \cdot 3^{\frac{1}{3}}}$$

$$\frac{3}{2^6 \cdot \sqrt[3]{3}}$$

$$\frac{3}{64} \cdot \frac{\sqrt[3]{2}}{\sqrt[3]{3^2}} \\ \frac{3 \sqrt[3]{2}}{32 \sqrt[3]{3^2}} \\ \frac{3 \sqrt[3]{3^2}}{32 \sqrt[3]{3^2}} \\ \frac{3}{32}$$

$$R + 0: \frac{3}{96}$$

$$r) \left(1 - \frac{2}{3}\right) \cdot \left(\frac{2}{3} - \frac{3}{4}\right)^2$$

$$\left(\frac{2}{3} - 1\right)^2 \cdot \left(\frac{1}{3} - 1\right)$$

$$\left(\frac{2 - 3}{3}\right) \cdot \left(\frac{8 - 9}{12}\right)^2$$

$$\left(\frac{2 - 6}{3}\right)^2 \cdot \left(\frac{7 - 3}{3}\right)$$

$$\left(-\frac{4}{3}\right)^2 \cdot \left(-\frac{1}{12}\right)$$

$$\left(-\frac{16}{9}\right) \cdot \left(-\frac{2}{3}\right)$$

$$\left(-\frac{16}{9}\right) \cdot \left(\frac{1}{12}\right)$$

$$\left(\frac{16}{9}\right) \cdot \left(-\frac{2}{3}\right)$$

$$\frac{-1}{288} \\ \frac{-32}{288}$$

$$R + 0: \frac{3}{1024}$$

$$5) \left(1; \frac{3}{2}\right) + \left(\frac{2}{3}; \frac{3}{4}\right)^2$$

$$\left(\frac{2}{3}; 2\right)^2 - \left(\frac{1}{3}; 1\right)$$

$$\left(\frac{2}{3}\right) + \left(\frac{8}{12}\right)^2$$

$$\left(\frac{2}{3}\right)^2 - \left(\frac{1}{3}\right)$$

$$\frac{2}{3} + \frac{1}{4}$$

$$\frac{1}{4} - \left(\frac{1}{3}\right)$$

$$\frac{7}{12}$$

$$R_{10} = -\frac{33}{8}$$

$$2) \frac{2}{\sqrt{6}} = \frac{2}{\sqrt{6}} \cdot \frac{\sqrt{6}}{\sqrt{6}} = \frac{2\sqrt{6}}{6} = \frac{\sqrt{6}}{3}$$

$$b) \frac{2}{3\sqrt{2}} = \frac{2}{3\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{2\sqrt{2}}{6} = \frac{\sqrt{2}}{3}$$

$$c) \frac{3}{\sqrt{7}} = \frac{3}{\sqrt{7}} \cdot \frac{\sqrt{7}}{\sqrt{7}} = \frac{3\sqrt{7}}{7}$$

$$d) \frac{2}{4-2\sqrt{3}} = \frac{2}{2(2-\sqrt{3})} = \frac{1}{2-\sqrt{3}} \cdot \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{2+\sqrt{3}}{4-3} = 2+\sqrt{3}$$

$$e) \frac{1}{1-\sqrt{5}} = \frac{1}{1-\sqrt{5}} \cdot \frac{1+\sqrt{5}}{1+\sqrt{5}} = \frac{1+\sqrt{5}}{1-5} = \frac{1+\sqrt{5}}{-4}$$

$$f) \frac{1}{\sqrt{5}-\sqrt{7}} = \frac{1}{\sqrt{5}-\sqrt{7}} \cdot \frac{\sqrt{5}+\sqrt{7}}{\sqrt{5}+\sqrt{7}} = \frac{\sqrt{5}+\sqrt{7}}{5-7} = \frac{\sqrt{5}+\sqrt{7}}{-2}$$

22) Justificación

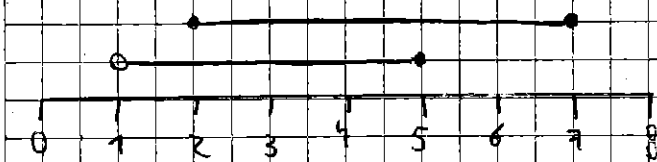
El valor absoluto de -8 es 8 $|-8| = 8$

La distancia entre -2 y 5 es 7 $|-2-5| = 7$

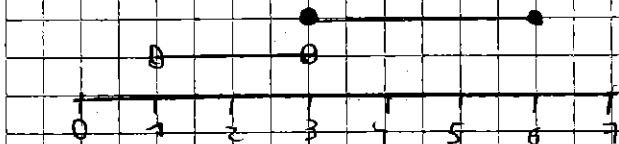
La distancia de $+3$ a 3 no es igual a $|+3|^2$ $|+3-3| \neq |-3|^2$
 $6 \neq 9$

La distancia de -5 a 5 es igual a 10 $|-5-5| = 10$

23) $(1;5] \cap [2;7] = [2;5]$



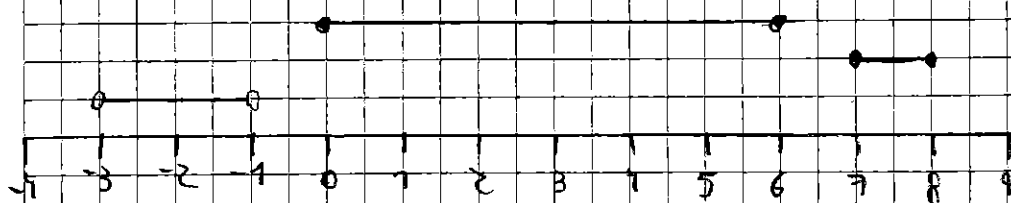
6) $(1;3) \cap [3;6] = [3;6]$



7) $(-1;3] \cap [0;3] = [-1;3]$



8) $\{(-3;-1) \cup [7;8]\} \cap [0;6] = \emptyset$



24)

a) $A = (5, 9]$

b) $B = [-7, 3]$

c) $C = (-\infty, -2) \cup (2, \infty)$

d) $D = (-9, -7) \cup (-7, 2)$

25)

a) $(-2, 3) = \{x \mid -2 < x < 3\}$

b) $[-1, 7] = \{x \mid -1 \leq x \leq 7\}$

c) $[5, \infty) = \{x \mid 5 \leq x\}$

d) $(-\infty, -5] = \{x \mid x \leq -5\}$

e) $(-\infty, \infty) = \{x \mid -\infty < x < \infty\}$

27) $\log 0,02 = x$

$$\log 2 \times 10^{-2} = \log 2 + \log 10^{-2}$$

$$\log 2 + -2 \log 10$$

$$0,3010 + 2 = -2,3010$$

b) $\log 5 = x$

c) $\log 0,0625 = x$

$$\log \frac{10}{2}$$

$$\log 625 \cdot 10^{-4}$$

$$\log 10 - \log 2 = -1,3010$$

$$\log 5 + \log 10^{-4}$$

$$4 \log \frac{10}{2} - 4$$

$$4 \log 2 - 4$$

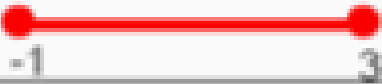
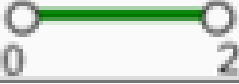
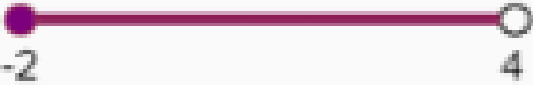
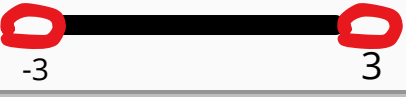
$$-2,776$$

d) $\log \sqrt[4]{8} = x$

$$\log 2^{\frac{3}{4}}$$

$$\frac{3}{4} \log 2$$

$$0,22575$$

	$[-1,3]$	$\{x \in \mathbb{R} / -1 \leq x \leq 3\}$
	$(0,2)$	$\{x \in \mathbb{R} / -1 < x < 3\}$
	$[-2,4)$	$\{x \in \mathbb{R} / -2 \leq x < 4\}$
	$[-2,1)$	$\{x \in \mathbb{R} / -2 \leq x < 1\}$
	$(1,5]$	$\{x \in \mathbb{R} / 1 < x \leq 5\}$
	$[-1,\infty)$	$\{x \in \mathbb{R} / -1 < x\}$
	$(-\infty,2)$	$\{x \in \mathbb{R} / x < 2\}$
	$(0,\infty)$	$\{x \in \mathbb{R} / 0 < x\}$
	$(-\infty,3]$	$\{x \in \mathbb{R} / x \leq 3\}$
	$(-1,5)$	$\{x \in \mathbb{R} / -1 < x < 5\}$
	$(-\infty,0]$	$\{x \in \mathbb{R} / x \leq 0\}$
	$[2/3,\infty)$	$\{x \in \mathbb{R} / (2/3) \leq x\}$
	$[-2,2]$	$\{x \in \mathbb{R} / -2 \leq x \leq 2\}$
	$(-3,3)$	$\{x \in \mathbb{R} / x < 3\}$
	$(-\infty,-3] \cup [3,\infty)$	$\{x \in \mathbb{R} / x \geq 3\}$
	$[2,\infty)$	$\{x \in \mathbb{R} / 2 \leq x\}$

$$28) \log_{1/2} 0,25 = y$$

$$\left(\frac{1}{2}\right)^y = 0,25 \quad \left(\frac{1}{2}\right)^y = \frac{1}{2^2} \quad y = 2$$

$$b) \log_{\sqrt{5}} 125 = y$$

$$(\sqrt{5})^y = 125 \quad \sqrt{5}^y = \sqrt{5^6} \quad y = 6$$

$$c) \log 0,001 = y$$

$$10^y = 0,001 \quad 10^y = 10^{-3} \quad y = -3$$

$$d) \log 10000 = y$$

$$10^y = 10000 \quad 10^y = 10^4 \quad y = 4$$

$$e) \ln \frac{1}{e^5}$$

$$e^y = \frac{1}{e^5} \quad e^y = e^{-5} \quad y = -5$$

$$f) \log_{\sqrt{3}} \sqrt{\frac{1}{81}} = y$$

$$\sqrt{3}^y = \sqrt{\frac{1}{81}}$$

$$\sqrt{3}^y = \sqrt{3^{-4}} \quad y = -4$$

$$g) \log_5 81 = -4$$

$$5^{-4} = 81$$

$$\frac{1}{5^4} = 81$$

$$\frac{1}{5^4} = \sqrt[4]{81}$$

$$5 = \sqrt[4]{81}$$

$$h) \log_2 y^3 = 6 \quad i) \log_2 \sqrt[5]{2} + \log_2 8 + \log_2 \frac{1}{2} = \frac{17}{5}$$

$$2^6 = y^3$$

$$4 = y$$

$$\frac{1}{5} \log_2 2 + 3 \log_2 2 + 1 \log_2 2$$

$$\frac{1}{5} + 3 + 1$$

$$\frac{11}{5}$$

$$j) \ln 1 + \ln e + \ln e^3 + \ln \sqrt[3]{e} + \ln \frac{1}{e} = \frac{10}{3}$$

$$0 + 1 + 3 + \frac{1}{3} - 1$$

$$\frac{10}{3}$$

$$K) \log 810 + \log 0.2^3 + \log 5 \sqrt{\frac{1}{4}}$$

$$\log 2 \cdot 3 \cdot 5 + \log 2 \cdot 10^{-2} + \frac{1}{2} \log 4$$

$$\log 3^4 + \log 10 + \log 3 + 2 \log 10 - \frac{2}{5} \log 3$$

$$4 \log 3 + 1 + \log 3 - 2 \frac{2}{5} \log 3$$

$$\frac{23}{5} \log 3 - 1 = \log \sqrt[5]{5^{23}} - 1 = \log 3 \sqrt[5]{3}$$

$$l) \log \sqrt[5]{0.01} + \log 3 \sqrt[3]{0.25} + \log \sqrt{\frac{16}{5}} =$$

$$\frac{1}{5} \log 4 - \frac{2}{5} \log 10 + \frac{1}{3} \log 25 - \frac{2}{3} \log 10 - \frac{1}{3} \log 8 + \frac{1}{2} \log 16 - \frac{1}{2} \log 10 - \frac{1}{2} \log 5$$

$$\frac{2}{5} \log 2 - \frac{2}{5} + \frac{2}{3} \log 5 - \frac{2}{3} - 1 \log 2 + 2 \log 2 - \frac{1}{2} - \frac{1}{2} \log 5$$

$$\frac{7}{5} \log 2 + \frac{1}{6} \log 3 - \frac{47}{30}$$

$$\log \sqrt[5]{2} + \log \sqrt[6]{5} = \frac{47}{30}$$

$$\log \sqrt[5]{2^7}, \sqrt[6]{5} = \frac{47}{30}$$

$$\log \sqrt[30]{2^{42} \cdot 5^3} = \frac{47}{30}$$

$$m) \log_2 \sqrt[5]{a} + \log_2 \frac{\sqrt[3]{a}}{\sqrt[2]{a}} + \log_2 \frac{1}{4}$$

$$\log_2 a^{\frac{1}{5}} + \log_2 \frac{a^{\frac{1}{3}}}{a^{\frac{1}{2}}} + \log_2 4^{-1}$$

$$\frac{1}{5} \log_2 a + \log_2 \frac{a^{\frac{1}{3}}}{a^{\frac{1}{2}}} + 1 \cdot 2$$

$$\frac{1}{5} + \frac{1}{6} - 2$$

$$-\frac{19}{30}$$

$$n) \log_{a-b} \sqrt[3]{\frac{a}{a-b}} + \log_{\frac{a}{b}} \frac{b}{a} + \log_{a+b} \sqrt{a+b}$$

$$\log_{a-b} (a-b)^{-\frac{1}{3}} + \log_{\frac{a}{b}} \frac{a^{-1}}{b} + \log_{a+b} (a+b)^{\frac{1}{2}}$$

$$-\frac{1}{3} - 1 + \frac{1}{2}$$

$$-\frac{5}{6}$$

$$o) \log_a \sqrt[3]{\frac{a}{b}} - \log_{\frac{a}{b}} \sqrt[5]{\frac{a}{b}} + \log_{ab} (ab)^{-3}$$

$$\log_a a^{\frac{10}{3}} - \frac{1}{5} \log_{\frac{a}{b}} b - 3 \log_{ab} ab$$

$$\frac{10}{3} - \frac{1}{5} - 3$$

$$\frac{2}{15}$$

$$29) \log(xy) - 2 \log\left(\frac{x}{y}\right)$$

$$\log(xy) - \log\left(\frac{x}{y}\right)^2$$

$$\log(xy)$$

$$\frac{x^2}{y^2}$$

$$\log(x^3 y^3)$$

$$\frac{x^2}{y^2}$$

$$\log\left(\frac{y^3}{x}\right)$$

$$0 < x$$

$$0 < y$$

$$b) 2 \ln(a-b) - \ln(a^2 - b^2)$$

$$\ln(a+b)^2 - \ln(a^2 - b^2)$$

$$\ln\left(\frac{(a-b) \cdot (a-b)}{(a-b) \cdot (a+b)}\right)$$

$$\ln\left(\frac{a-b}{a+b}\right)$$

$$c) 4 \log_2 \frac{\sqrt{a-b}}{a} - \frac{1}{2} \log_2 \left(\frac{a-b}{a}\right)^4$$

$$2 \log_2 \frac{a-b}{a} - \frac{1}{2} \log_2 \left(\frac{a-b}{a}\right)^4$$

0

$$3d) \ln \frac{x^2 y (m+n)}{mn}$$

$$\ln \left(\frac{x^2 y m + x^2 y n}{mn} \right)$$

$$\ln(x^2) + \ln(y) + \ln(m+n) + \ln(m) - (\ln(n))$$

$$2 \ln(x) + \ln(y) + \ln(m+n) + \ln(m) - \ln(n)$$

$$e) \log_2 \frac{a^2 - b^2}{a \cdot b}$$

$$\log_2 \frac{(a-b) \cdot (a+b)}{a \cdot b}$$

$$\log_2(a-b) + \log_2(a+b) - \log_2(a) - \log_2(b)$$

$$f) \log_2 \sqrt[3]{2\sqrt{2} \sqrt{2}}$$

$$\log_2 \frac{2}{2} =$$

$$\frac{1}{6} \log_2 2$$

Números complexos

$$3) (3+6i) + (2-8i) = 5 + 3i$$

$$6) (7+5i) - (1+2i) = 6 + 3i$$

$$c) (5+9i) \cdot (3-7i) = 15 - 20i + 27i + 28$$

$$+ 3 + 7i$$

$$d) (2-3i) : (-1+4i) = \frac{2-3i}{-1+4i} \cdot \frac{-1-4i}{-1-4i}$$

$$\frac{(2-3i) \cdot (-1-4i)}{(-1+4i) \cdot (-1-4i)}$$

$$\frac{-2-7i-3i-12}{1+16}$$

$$\frac{-14-10i}{17}$$

$$\frac{-14}{17} - \frac{10}{17}i$$

$$e) (2-3i) : (-1+4i) \cdot (1-7i) \quad f) (3+6i)^2$$

$$\frac{(2-3i) \cdot (1-7i)}{-1+4i}$$

$$\frac{(-10-7i)}{-1+4i}$$

$$\frac{(-10-7i) \cdot (-1-4i)}{(-1+4i) \cdot (-1-4i)}$$

$$\frac{-34+57i}{17}$$

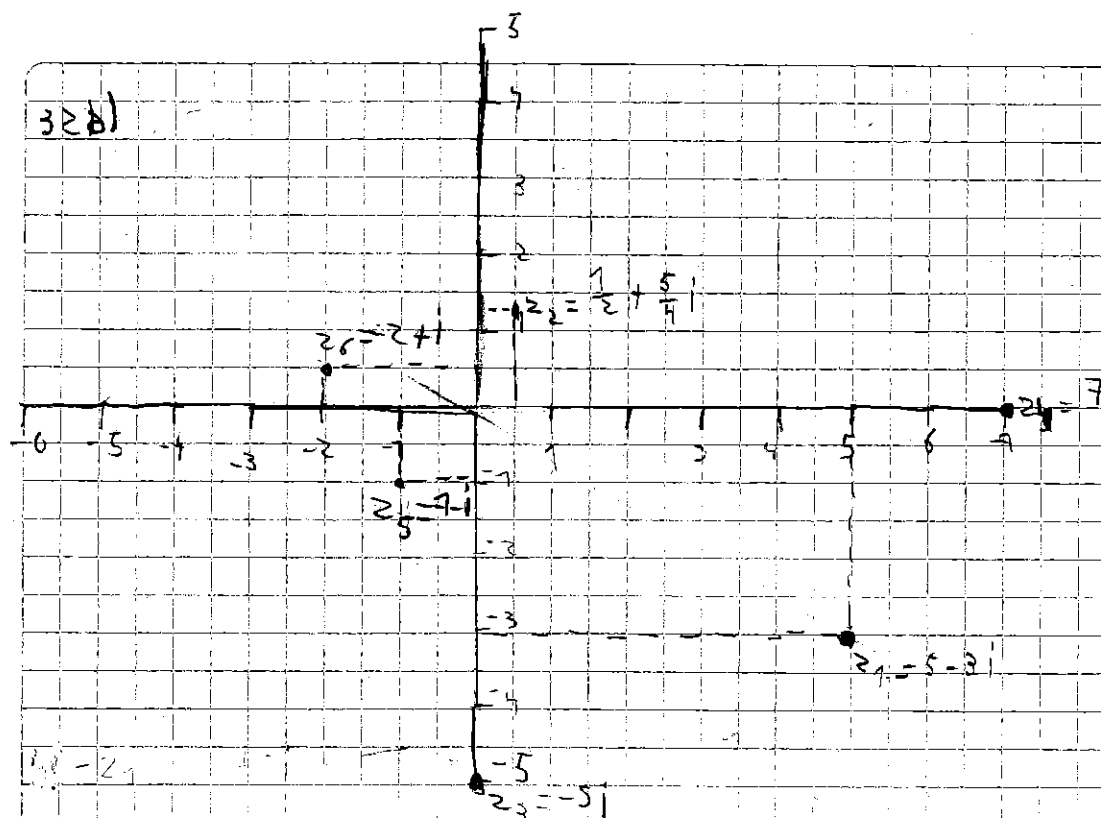
$$-2 + 3i$$

$$(3+6i) \cdot (3+6i)$$

$$9 + 18i + 18i + 36$$

$$+ 27 + 36i$$

328)



$$b) \quad z_1 = 5 - 3i \quad \bar{z}_1 = 5 + 3i \quad z_2 = \frac{1}{2} + \frac{5}{4}i \quad \bar{z}_2 = \frac{1}{2} - \frac{5}{4}i$$

$$z_3 = 5i \quad \bar{z}_3 = -5i \quad z_4 = -7 \quad \bar{z}_4 = -7 \quad z_5 = -1 - i \quad \bar{z}_5 = -1 + i$$

$$z_6 = -2 + i \quad \bar{z}_6 = -2 - i$$

$$c) \quad z_1 z_2 = (5 - 3i) \cdot \left(\frac{1}{2} + \frac{5}{4}i\right) = \frac{5}{2} + \frac{25}{4}i - \frac{3}{2}i + \frac{15}{4} = \frac{25}{4} + \frac{19}{4}i$$

$$i) \quad z_1 z_5 = (5 - 3i) \cdot (-1 - i) = -1 + 4i$$

$$ii) \quad z_2 z_5 = \left(\frac{1}{2} + \frac{5}{4}i\right) \cdot (-1 - i) = -\frac{3}{2} + \frac{9}{4}i$$

$$iii) \quad z_6 z_6 = (-2 + i) \cdot (-2 - i) = 5$$

$$iv) \quad z_5 z_6 = (-1 - i) \cdot (-2 + i) = \frac{1}{5} + \frac{3}{5}i$$

$$v) \quad z_6 z_3 = (-2 + i) \cdot (-5i) = -\frac{1}{5} - \frac{2}{5}i$$

$$33) (3-6i) \cdot (4+ki) = (12+3ki) - 24i + 6k$$

$$(12+6k) + (3ki-24i) \quad k=-2$$

$$(12-12) + (-6i-24i)$$

$$0 - 30i$$

$$34) (3-6i) \cdot (4+ki) = (12+6ki) + (3ki-24i) \quad k=8$$

$$(12+48) + (24i-24i)$$

$$60 + 0i$$

$$34) (2+xi)^2 = (2+xi) \cdot (2+xi) = 4 + 2xi + 2xi - x$$

$$(4-x) + (2(2xi)) \quad x \neq 4$$

$$0 + 16i$$

$$35) 2(3+i) - (2-4i) \cdot (2-4i) \quad 6) (1+2i)^2 + (-3+4i)i - (2-2i) \cdot (1-2i)$$

$$(6+2i) - (4-8i-8-16) \quad (1+2i)(1+2i) + (-3i-4) - (2-4i-2i-4)$$

$$6+2i + 12 + 16i \quad (1+2i+2i-4) + (-3i-4) - (-2-6i)$$

$$18 + 18i \quad (-3+4i) - 3i - 4 + 2 + 6i$$

$$c) (-1+\frac{1}{2}i) \cdot (1+\frac{1}{2}i) + (\frac{1}{2}+2i) - (\frac{1}{3}-\frac{1}{2}i)i$$

$$-\frac{5}{4} + \frac{1}{2} + 2i - \frac{1}{2} - \frac{1}{3}i$$

$$-\frac{5}{4} + \frac{5}{3}i$$

$$d) \frac{(3-4i)(2+i)}{(2-3i)(1-2i)}$$

$$\frac{10-5i}{-3-4i}$$

$$\frac{10-5i}{-3-4i} \cdot \frac{-3+4i}{-3+4i}$$

$$\frac{(10-5i)(-3+4i)}{(-3-4i)(-3+4i)}$$

$$\frac{-10+55i}{25}$$

$$= \frac{-2}{5} + \frac{11i}{5}$$

$$e) \frac{(1-i)(2+i)}{(2-3i)-2(1-2i)}$$

$$\frac{3-i}{-1-3i}$$

$$\frac{3-i}{-1-3i} \cdot \frac{-1+i}{-1+i}$$

$$\frac{(3-i)(-1+i)}{(-1-3i)(-1+i)}$$

$$\frac{-1-3i}{-1-3i}$$

$$\frac{1}{\sqrt{(-1)^2 + (-3)^2}}$$

$$\frac{1}{\sqrt{10}}$$

361

$$a) (1+2i)(3-i)+(-2i) \quad b) z_1 \cdot \bar{z}_2 = (1+2i)(3+i)$$

$$(5+5i)+(-2i)$$

$$5+3i$$

$$1+7i$$

$$\bar{z}_1 \cdot z_2 = (1-2i)(3-i)$$

$$1-7i$$

$$z_1 \cdot \bar{z}_2 = \bar{z}_1 \cdot z_2 = 1+7i-7+7i$$

$$+7i$$

$$z_1 \bar{z}_2 - \bar{z}_1 z_2 = -14i$$

$$c) |(1+2i)^2 \cdot ((3-i)+(-2i))| \quad d) |z_1 \cdot ((3-i)+(-2i))|$$

$$|(-3+4i)(3-3i)|$$

$$|3+2i|$$

$$\sqrt{3^2+2^2}$$

$$5\sqrt{2}$$

$$|2i|$$

$$|2i \cdot (3-3i)|$$

$$|6+6i|$$

$$\frac{6\sqrt{2}}{2}$$

$$3\sqrt{2}$$

$$e) z_1, z_2 = (1+2i) \cdot (3-i)$$

$$= 5 + 5i$$

$$\overline{z_1 \cdot z_2} = 5 - 5i$$

$$\left| \frac{z_1 + (3-i)}{5-5i} \right|$$

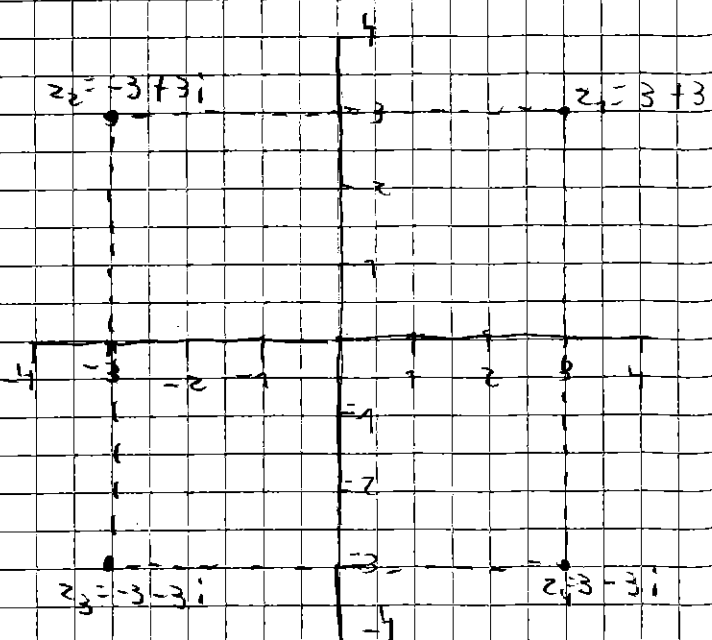
$$\left| \frac{3+i}{5-5i} \cdot \frac{5+5i}{5+5i} \right|$$

$$\left| \frac{10+20i}{50} \right|$$

$$\left| \frac{1}{5} + \frac{2i}{5} \right|$$

$$\frac{\sqrt{5}}{5}$$

37)



$$\text{Area} = 6.6 = 36$$

$$\text{perimeter} = 4.6 = 24$$